# Jun Zeng

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The latest update of CV was done by Jun Zeng on October 16, 2026.

## Research Experience

I am interested in researching mathematical foundations and developing modern algorithms for autonomous robots, achieving low-latency, agile, and safe maneuvers with adaptation and robustness in complex environments. My research lies in the combination of control, optimization, motion planning, trajectory generation, reinforcement learning, and machine learning.

## Experience

**Chief Technology Officer** Oct 2025 – Present
*Cyan Robotics* *Shanghai, China*

- Technical Leadership: Own technology strategy, system architecture, and engineering roadmap. Grew engineering from ∼15 to ∼40 across mechanical and electrical design, embedded systems, robot control and whole-body autonomy, and multimodal end-to-end models, keeping hardware and software teams on a single roadmap.
- Dino OS: Proposed and architected a modern embodied AI framework unifying build, test, training, fine-tuning and deployment across heterogeneous compute platforms. One reproducible toolchain from experiment to on-robot deployment, and the pivotal foundation for a fast-moving, complex embodied program; deployed on Amoo.
- Hardware-Software Co-Design: Directed in-house IMU, stereo dToF module, and modular robot joints. Vertical integration cut BOM cost and removed vendor lock-in, while native driver integration enabled cross-sensor time synchronization and low-level control unavailable off the shelf.
- Product & Business Impact: Took Amoo from research prototype to integrated consumer product. Global debut at AWE 2026, followed by various venues such as BEYOND Expo 2026 in Macao, WAIC in Shanghai, WRC in Beijing, IFA in Berlin; over 1000 letters of intent to purchase.

**Senior AI/ML Scientist** Jun 2022 – Oct 2025
*Cruise LLC* *San Francisco, CA*

- Large Behavior Models & AI Guardrails: Worked as one of core ICs for end-to-end large behavior model experience spanning pretraining, post-training, evaluation harness, and deployment. Designed and improved auto-regressive trajectory decoder, enhancing dynamic feasibility of the generated trajectories from the large ML model. Designed and contributed improvements to trajectory constraints enforcement in both training and inference time using various techniques such as flow matching, adapter layers fine-tuning, etc. Contributed to improve algorithm for training stability and robustness when transitioning between manual and autonomous driving, ensuring better policy learning and deployment.
- Trajectory Generation: Led the unification of reactive motion planning and control to enhance real-time adaptability and reduce system complexity. Led next generation of coupled non-convex trajectory generation solver with latest techniques of inner point methods. Designed and developed ML-based warm start calculation to accelerate solver runtime. Designed and integrated a ML-based dragging detector with collision imminence.
- Cross Team Technical Leadership: Worked as one of core ICs to reduce unnecessary braking and swerving to improve company-level top-down metrics; Collaborated closely with prediction and behavior planning teams to refine trajectory generation for complex driving scenarios, such as tight turns and cautious VRU interactions, contributing to more robust large planner model training and ML-based trajectory selection.

## Education

**University of California, Berkeley** Aug 2017 - May 2022
Ph.D. in Mechanical Engineering
Advisor: Prof. Koushil Sreenath
Field: Control, Robotics, Machine Learning

**Ecole Polytechnique** Mar 2015 - Aug 2017
Diplôme d'ingénieur in Applied Mathematics and Mechanics

**Shanghai Jiao Tong University** Sep 2012 - May 2016
Bachelor of Science and Engineering in Automation

## Honors & Awards

- **Chin Leung Shui Chun Fellowship**, UC Berkeley *2021*
- **Graduate Division Block Fellowship**, UC Berkeley *2020 - 2021*
- **Mechanical Engineering Department Award**, UC Berkeley *2020*
- **X Fondation Fellowship**, Ecole Polytechnique *2015 - 2017*
- **Ardian Excellence Scholarship**, SJTU *2014*
- **University Excellence Scholarship**, SJTU *2012 - 2014*

## Publications

[1] Xuemin Chi*, **Jun Zeng**, Jihao Huang, Zhitao Liu, and Hongye Su. "Fast Path Planning for Autonomous Vehicle Parking with Safety-Guarantee using Hamilton-Jacobi Reachability". In: *IEEE Transactions on Vehicular Technology* (2026).

[2] Jihao Huang*, **Jun Zeng**, Xuemin Chi, Koushil Sreenath, Zhitao Liu, and Hongye Su. "Dynamic Collision Avoidance Using Velocity Obstacle-Based Control Barrier Functions". In: *IEEE Transactions on Control Systems Technology* 33.5 (2025), pp. 1601–1615.

[3] Shuo Liu*, Zhe Huang, **Jun Zeng**, Koushil Sreenath, and Calin A. Belta. "Learning-Enabled Iterative Convex Optimization for Safety-Critical Model Predictive Control". In: *IEEE Open Journal of Control Systems* 4 (2025), pp. 482–500.

[4] Bei Zhou*, Cheng Hu, **Jun Zeng**, Zhouheng Li, Johannes Betz, Lei Xie, and Hongye Su. "Adaptive learning-based model predictive control strategy for drift vehicles". In: *Robotics and Autonomous Systems* 188 (2025), p. 104941.

[5] Jihao Huang*, Zhitao Liu, **Jun Zeng**, Xuemin Chi, and Hongye Su. "Obstacle Avoidance for Unicycle-Modelled Mobile Robots with Time-varying Control Barrier Functions". In: *2023 49th Annual Conference of the IEEE Industrial Electronics Society (IECON)*. 2023.

[6] Jihao Huang*, **Jun Zeng***, Xuemin Chi, Koushil Sreenath, Zhitao Liu, and Hongye Su. "Velocity Obstacle for Polytopic Collision Avoidance for Distributed Multi-Robot Systems". In: *IEEE Robotics and Automation Letters* 8.6 (2023), pp. 3502–3509.

[7] Zhongyu Li, **Jun Zeng**, Shuxiao Chen, and Koushil Sreenath. "Autonomous navigation of underactuated bipedal robots in height-constrained environments". In: *The International Journal of Robotics Research* 42.8 (2023), pp. 565–585.

[8] Qiayuan Liao, Zhongyu Li, Akshay Thirugnanam, **Jun Zeng**, and Koushil Sreenath. "Walking in Narrow Spaces: Safety-critical Locomotion Control for Quadrupedal Robots with Duality-based Optimization". In: *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2023, pp. 2723–2730.

[9] Shuo Liu*, **Jun Zeng***, Koushil Sreenath, and Calin A Belta. "Iterative Convex Optimization for Model Predictive Control with Discrete-Time High-Order Control Barrier Functions". In: *2023 American Control Conference (ACC)*. 2023, pp. 3368–3375.

[10] Yifan Zeng*, Suiyi He*, Han Hoang Nguyen, Yihan Li, Zhongyu Li, Koushil Sreenath, and **Jun Zeng**. "i2LQR: Iterative LQR for Iterative Tasks in Dynamic Environments". In: *2023 62nd IEEE Conference on Decision and Control (CDC)*. 2023, pp. 5255–5260.

[11] Suiyi He*, **Jun Zeng***, and Koushil Sreenath. "Autonomous Racing with Multiple Vehicles using a Parallelized Optimization with Safety Guarantee using Control Barrier Functions". In: *2022 International Conference on Robotics and Automation (ICRA)*. 2022, pp. 3444–3451.

[12] Ashish Kumar*, Zhongyu Li*, **Jun Zeng**, Deepak Pathak, Koushil Sreenath, and Jitendra Malik. "Adapting Rapid Motor Adaptation for Bipedal Robots". In: *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2022, pp. 1161–1168.

[13] Zhongyu Li, **Jun Zeng**, Akshay Thirugnanam, and Koushil Sreenath. "Bridging Model-based Safety and Model-free Reinforcement Learning through System Identification of Low Dimensional Linear Models". In: *Robotics: Science and Systems (RSS)*. 2022.

[14] Akshay Thirugnanam, **Jun Zeng**, and Koushil Sreenath. "Duality-based Convex Optimization for Real-time Obstacle Avoidance between Polytopes with Control Barrier Functions". In: *2022 American Control Conference (ACC)*. 2022, pp. 2239–2246.

[15] Akshay Thirugnanam*, **Jun Zeng***, and Koushil Sreenath. "Safety-Critical Control and Planning for Obstacle Avoidance between Polytopes with Control Barrier Functions". In: *2022 International Conference on Robotics and Automation (ICRA)*. 2022, pp. 286–292.

[16] Xiangyu Wu, **Jun Zeng**, Andrea Tagliabue, and Mark W. Mueller. "Model-Free Online Motion Adaptation for Energy-Efficient Flight of Multicopters". In: *IEEE Access* 10 (2022), pp. 65507–65519.

[17] Chenyu Yang*, Guo Ning Sue*, Zhongyu Li*, Lizhi Yang, Haotian Shen, Yufeng Chi, Akshara Rai, **Jun Zeng**, and Koushil Sreenath. "Collaborative Navigation and Manipulation of a Cable-Towed Load by Multiple Quadrupedal Robots". In: *IEEE Robotics and Automation Letters* 7.4 (2022), pp. 10041–10048.

[18] Lizhi Yang*, Zhongyu Li*, **Jun Zeng**, and Koushil Sreenath. "Bayesian Optimization Meets Hybrid Zero Dynamics: Safe Parameter Learning for Bipedal Locomotion Control". In: *2022 International Conference on Robotics and Automation (ICRA)*. 2022, pp. 10456–10462.

[19] Scott Gilroy*, Derek Lau*, Lizhi Yang, Ed Izaguirre, Kristen Biermayer, Anxing Xiao, Mengti Sun, Ayush Agrawal, **Jun Zeng**, Zhongyu Li, and Koushil Sreenath. "Autonomous Navigation for Quadrupedal Robots with Optimized Jumping through Constrained Obstacles". In: *2021 IEEE 17th International Conference on Automation Science and Engineering (CASE)*. 2021, pp. 2132–2139.

[20] Suiyi He, **Jun Zeng**, Bike Zhang, and Koushil Sreenath. "Rule-Based Safety-Critical Control Design using Control Barrier Functions with Application to Autonomous Lane Change". In: *2021 American Control Conference (ACC)*. 2021, pp. 178–185.

[21] Anxing Xiao*, Wenzhe Tong*, Lizhi Yang*, **Jun Zeng**, Zhongyu Li, and Koushil Sreenath. "Robotic Guide Dog: Leading a Human with Leash-Guided Hybrid Physical Interaction". In: *2021 IEEE International Conference on Robotics and Automation (ICRA)*. 2021, pp. 11470–11476.

[22] **Jun Zeng***, Zhongyu Li*, and Koushil Sreenath. "Enhancing Feasibility and Safety of Nonlinear Model Predictive Control with Discrete-Time Control Barrier Functions". In: *2021 60th IEEE Conference on Decision and Control (CDC)*. 2021, pp. 6137–6144.

[23] **Jun Zeng***, Bike Zhang, Zhongyu Li, and Koushil Sreenath. "Safety-Critical Control using Optimal-decay Control Barrier Function with Guaranteed Point-wise Feasibility". In: *2021 American Control Conference (ACC)*. 2021, pp. 3856–3863.

[24] **Jun Zeng***, Bike Zhang*, and Koushil Sreenath. "Safety-critical model predictive control with discrete-time control barrier function". In: *2021 American Control Conference (ACC)*. 2021, pp. 3882–3889.

[25] Chenyu Yang*, Bike Zhang, **Jun Zeng**, Ayush Agrawal, and Koushil Sreenath. "Dynamic legged manipulation of a ball through multi-contact optimization". In: *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2020, pp. 7513–7520.

[26] **Jun Zeng***, Prasanth Kotaru, Mark W Mueller, and Koushil Sreenath. "Differential flatness based path planning with direct collocation on hybrid modes for a quadrotor with a cable-suspended payload". In: *IEEE Robotics and Automation Letters (RA-L)* 5.2 (2020), pp. 3074–3081.

[27] **Jun Zeng***, Prasanth Kotaru, and Koushil Sreenath. "Geometric Control and Differential Flatness of a Quadrotor UAV with Load Suspended from a Pulley". In: *2019 American Control Conference (ACC)*. 2019, pp. 2420–2427.

[28] **Jun Zeng*** and Koushil Sreenath. "Geometric Control of a Quadrotor with a Load Suspended from an Offset". In: *2019 American Control Conference (ACC)*. 2019, pp. 3044–3050.

## Open-Source Tools & Software

**Car-Racing: A toolkit to validate algorithm for Autonomous Car Racing**
- Versatile and Scalable simulation platform for autonomous car racing problems supporting dynamics, control, planning algorithm.
- Code repository: *https://github.com/HybridRobotics/car-racing*

**Python function for APIs for control barrier functions in the field of control, planning and navigation**
- A python API library of multiple algorithm of control barrier functions for control and robotics applications.
- Code repository: *https://github.com/HybridRobotics/cbf*

**Code Collection of Discrete-time Control Barrier Functions (DCBF)**
- A collection of multiple algorithm of discrete-time control barrier functions for control and robotics applications. This collection makes it simple to track latest work about control barrier functions.
- Code repository: *https://github.com/HybridRobotics/NMPC-DCLF-DCBF*

## Invited Talks

**Safety-Critical Model Predictive Control with Discrete-Time Control Barrier Function**
*3rd Norcal Control Workshop, Stanford University (Virtual), CA* Jan 2021

## Professional Service

**Program Committee**

- AAAI-23 special track on Safe and Robust AI *2023*
- ICRA-2022 Workshop on Opportunities and Challenges with Autonomous Racing (OCAR-ICRA) *2022*

**Journal Reviewers**

- ASME Journal of Dynamic Systems, Measurement, and Control (J. Dyn. Sys., Meas., Control.) *2021 - 2025*
- Automatica *2022 - 2026*
- Control Engineering Practice *2025 - 2026*
- IEEE Access *2022 - 2025*
- IEEE Control Systems Letters (L-CSS) *2023 - 2027*
- IEEE Open Journal of Control Systems (OJCSYS) *2023 - 2025*
- IEEE Robotics and Automation Letters (RA-L) *2020 - 2026*
- IEEE Transactions on Automatic Control (TAC) *2021 - 2026*
- IEEE Transactions on Automation Science and Engineering (T-ASE) *2022 - 2026*
- IEEE Transactions on Control of Network Systems (T-CONES) *2021 - 2025*
- IEEE Transactions on Control Systems Technology (T-CST) *2021 - 2026*
- IEEE Transactions on Industrial Electronics (TIE) *2024 - 2027*
- IEEE Transactions on Intelligent Transportation Systems (T-ITS) *2024 - 2026*
- IEEE Transactions on Robotics (TR-O) *2023 - 2026*
- IEEE Transactions on Systems, Man, and Cybernetics (TSMC) *2026*
- IEEE Transactions on Vehicular Technology (T-VT) *2025 - 2027*
- IEEE/ASME Transactions on Mechatronics (T-Mech) *2022 - 2027*
- International Journal of Robotics Research (IJRR) *2023 - 2026*
- International Journal of Robust and Nonlinear Control (RNC) *2023 - 2026*

**Conference Reviewers**

- AAAI Conference on Artificial Intelligence (AAAI) *2022 - 2024*
- American Control Conference (ACC) *2023 - 2027*
- Annual Conference of the IEEE Industrial Electronics Society (IECON) *2023 - 2025*
- European Control Conference (ECC) *2023 - 2025*
- IEEE Conference on Control Technology and Applications (CCTA) *2024 - 2025*
- IEEE Conference on Decision and Control (CDC) *2022 - 2026*
- IEEE International Conference on Robotics and Automation (ICRA) *2020 - 2027*
- IEEE International Conference on Unmanned Aircraft Systems (ICUAS) *2020 - 2021*
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) *2021 - 2025*
- Learning for Dynamics & Control Conference (L4DC) *2023 - 2026*

## Other Professional Experiences

**Waymo LLC** May 2021 - Aug 2021
Software Engineer Intern

**PlusAI** May 2019 - Aug 2019
Software Engineer Intern

**Valeo** May 2016 - Aug 2016
Summer Intern

**Autonomous Systems and Robotics Lab, ENSTA** Sep 2015 – Mar 2016
Research Assistant

## Teaching Experience

- **EECS 106B/206B: Robotic Manipulation and Interaction** 2020 Spring
  Graduate Student Instructor, UC Berkeley
- **EECS 106A/206A: Introduction to Robotics** 2019 Fall
  Graduate Student Instructor, UC Berkeley